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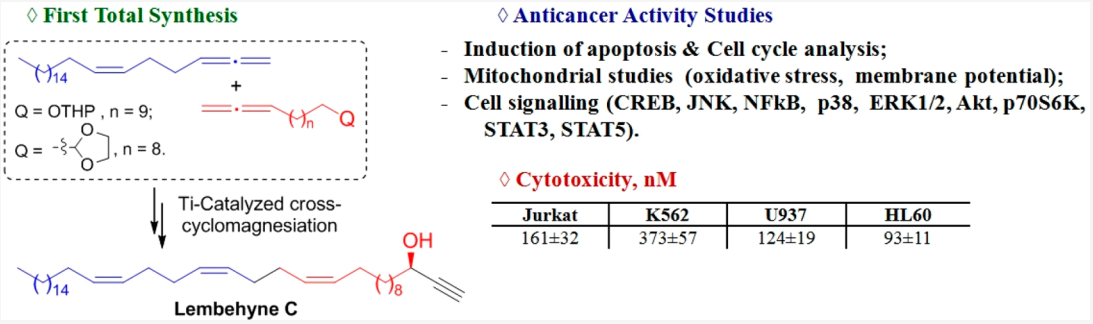
Total Synthesis of Natural Lembehyne C and Investigation of Its Cytotoxic Properties

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ABSTRACT: The first *Z*-stereoselective method for the synthesis of the natural marine alkynol lembehyne C, containing a 1*Z*,5*Z*,9*Z*-triene moiety, in 41% yield was developed using the new Ti-catalyzed cross-coupling of oxygenated and aliphatic 1,2-dienes as the key step. It was found for the first time that lembehyne C exhibits moderate cytotoxicity against Jurkat, K562, U937, and HL60 cancer cells and also efficiently induces apoptosis in Jurkat cells, with the cell death mechanism being activated by the mitochondrial pathway. The lembehyne C inhibition of the cell cycle follows the mitotic catastrophe mechanism.

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Currently, several thousand organic compounds occurring in marine organisms have been isolated and characterized. Most of these compounds exhibit anticancer, antibacterial, antiviral, antifungal, antiparasitic, or other types of biological activity.[1] The isolated compounds show pronounced structural diversity, including steroids, macrolides, lactams, alkaloids, nucleosides, fatty acids, and various long- and short-chain functionally substituted alkenes and alkynes. Most often, particular biologically active components are present in biomass in micro- or nanogram amounts; this is a significant obstacle to their detailed biological testing and fulfillment of their biomedical potential.

On the basis of the foregoing, one can state that studying the natural products isolated from marine species is an important area at the boundary of biology and chemistry. Synthesis and modification of these compounds and investigations of their effects on biological processes may lead to the development of new biologically active products, which, in turn, would give rise to new efficacious drugs.

Previously, it was shown that catalytic cross-cyclomagnesiation of oxygenated and aliphatic 1,2-dienes with Grignard reagents in the presence of Cp_2TiCl_2 (5−10 mol %) may serve as the key step for the six-step stereoselective total synthesis of a natural alkynol, lembehyne B (Figure 1). This, in turn, made it possible to produce lembehyne B in sufficient amounts to study its anticancer activity.[2]

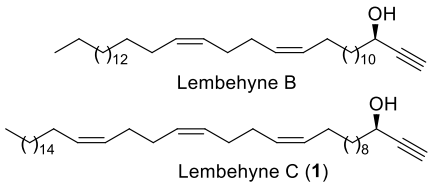


Figure 1. Natural acetylenic alcohols lembehynes B and C.

Herein, we developed a total synthesis of lembehyne C (Figure 1), an alkynol containing three bis-methylene-separated *Z*-double bonds, previously isolated in trace amounts from the Indonesian marine sponge *Haliclona* sp., with Ti-catalyzed cross-cyclomagnesiation of 1,2-dienes as the key step.[3] In view of the fact that lembehyne B exhibited anticancer activity against human leukemia cells by inducing early apoptosis, it seemed appropriate to produce a sufficient amount of lembehyne C in order to study its anticancer activity. It is noteworthy that previously Japanese researchers reported high neuritogenic

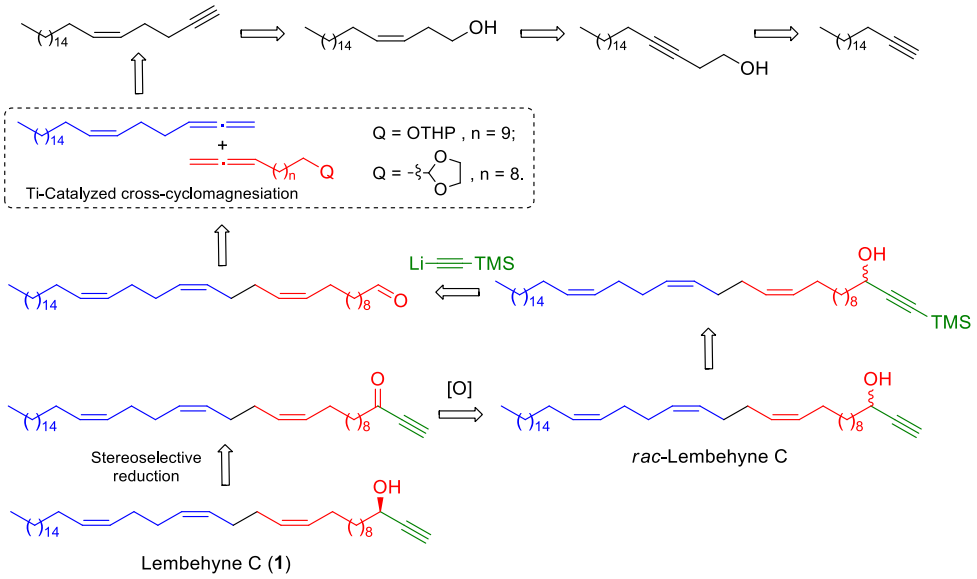
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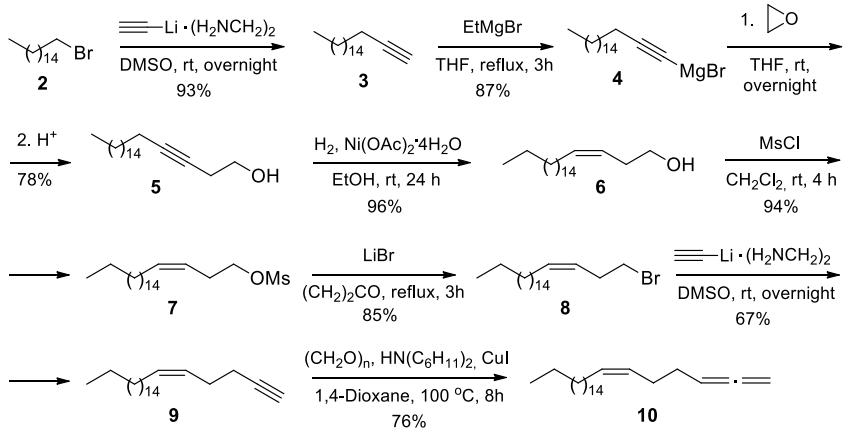
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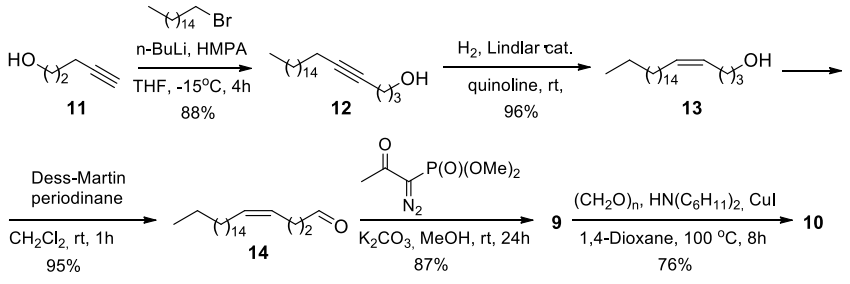
Scheme 1. Retrosynthetic Analysis of Lembehyne C



Scheme 2. Synthesis of (6*Z*)-Triacosa-1,2,6-triene (10)



Scheme 3. Alternative Approach to the Synthesis of (6*Z*)-Triacosa-1,2,6-triene (10)



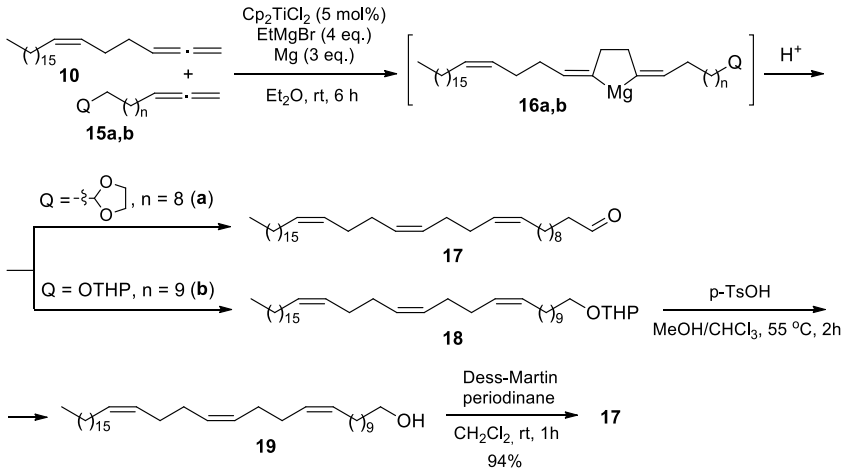
activity of lembehynes and have simultaneously ascertained that these compounds are efficient low-molecular-weight analogues of proteins of the neurotrophin family whose function is to support the viability of neurons.[4] Therefore, lembehynes are considered as potential pharmaceutical drugs for the treatment of neurodegenerative diseases such as Alzheimer's and Parkinson's diseases and Huntington's chorea.

RESULTS AND DISCUSSION

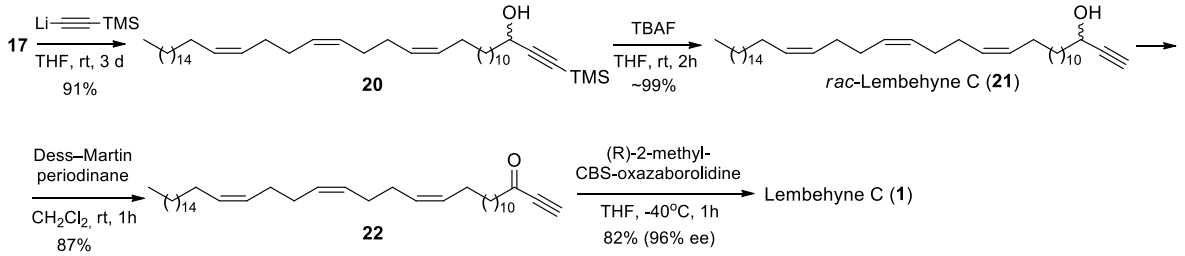
Chemistry. Despite the seeming simplicity of the structure of lembehyne C, its total synthesis is a challenging task because of the necessity to form three bis-methylene-separated *Z*-double bonds. No efficient methods for the formation of the 1*Z*,5*Z*,9*Z*-triene moiety in lembehyne C were reported, except for the interesting but laborious approach to unbranched 1,5,9,*n*-polyenes developed by A. Stark for the total synthesis of an acetogenin, chatenaytrienin-4.[5] To solve this problem, we resorted to the new Ti-catalyzed cross-cyclomagnesiation of functionally substituted and aliphatic 1,2-diynes, previously developed and which had been used for the total syntheses of lembehynes A and B, 5*Z*,9*Z*-dienoic acids, acetogenins, macrodiolides, and some insect pest pheromones.[6−10]

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Scheme 4. (Z,Z,Z)-Stereoselective Synthesis of Hexaconta-11Z,13Z,17Z-trienal (17)



Scheme 5. Total Synthesis of Natural Lembehyne C (1)



Preliminary retrosynthetic analysis of lembehyne C showed that the total synthesis can be accomplished by means of Ti-catalyzed cross-cyclomagnesiation of (6Z)-tricosa-1,2,6-triene with 11,12-tridecadienoltetrahydropyran ether or 2-dodeca-10,11-dien-1-yl-1,3- dioxolane (Scheme 1). The subsequent steps were intended to include generation of the terminal propargyl moiety to form racemic lembehyne C and oxidation of the carbinol group followed by stereoselective reduction (Scheme 1).

It is noteworthy that the synthesis of the key monomer (6Z)-tricosa-1,2,6-triene is a fairly complicated task; therefore, two approaches were developed for its synthesis from commercially available 1-bromohexadecane (**2**) (Scheme 2) or 4-pentyn-1-ol (**11**) (Scheme 3). The former approach comprising eight steps involves the reaction of 1-bromohexadecane (**2**) with the lithium acetylide ethylenediamine complex in DMSO to give 1-octadecyne (**3**) in 93% yield.[11] Compound **3** is treated with EtMgBr in THF at reflux for 3 h to give the corresponding magnesium acetylide **4**, which reacts with ethylene oxide and then undergoes acid hydrolysis to give eicos-3-yn-1-ol (**5**). Next, (3Z)-eicos-3-en-1-ol (**6**) was synthesized via $Ni(OAc)_2$-catalyzed hydrogenation of alkynol **5**; then alcohol **6** was treated with $MeSO_2Cl$ in CH_2Cl_2 for 4 h to give mesylate **7**, which was converted to 1-bromo-(3Z)-eicos-3-ene (**8**) in 85% yield by refluxing with LiBr in acetone for 3 h. The final stage included preparation of alkyne **9** by the reaction of alkenyl bromide **8** with lithium acetylide ethylenediamine complex and the subsequent acetylene−allene rearrangement of alkyne **9** under Mannich reaction conditions to afford the target (6Z)-tricosa-1,2,6-triene (**10**).

The alternative (five-step) method proposed for the synthesis of (6Z)-tricosa-1,2,6-triene (**10**) is based on 4-pentyl-1-ol (**11**) as the starting material. By successive treatment with *n*-BuLi and 1-bromohexadecane, this compound was converted to 4-heneicosyn-1-ol (**12**), which was subjected to catalytic hydrogenation with molecular hydrogen in the presence of $Ni(OAc)_2$ to afford 4Z-heneicos-4-en-1-ol (**13**) in 85% yield over two steps (Scheme 3). The Dess-Martin periodinane oxidation of alcohol **13** afforded 4Z-heneicos-4-en-1-al (**14**), which reacted with dimethyl (1-diazo-2-oxopropyl)phosphonate to afford (5Z)-docos-5-en-1-yne (**9**).[12] In the last step alkyne **9** underwent acetylene−allene rearrangement on refluxing with paraformaldehyde and dihexylamine in the presence of CuI in dioxane (100 °C, 8 h) to afford allene **10** (Scheme 3).

The next stage of the synthesis involved the intermolecular cross-cyclomagnesiation of (6Z)-tricosa-1,2,6-triene (**10**) and 2-dodeca-10,11-dien-1-yl-1,3-dioxolane (**15a**) or 11,12-tridecadienol tetrahydropyran ether (**15b**) with EtMgBr in the presence of Mg (powder) and catalytic amounts of Cp_2TiCl_2 [**10**:**15a** (**15b**):EtMgBr:Mg:[Ti] = 12:10:36:24:0.1, Et_2O, 20−22 °C, 10 h]. The acid hydrolysis of magnesacyclopentanes **16a,b** formed in situ afforded aldehyde **17** and tetrahydropyranyl ether **18** containing 1Z,5Z,9Z-diene moieties in 81% and 89% yields, respectively (Scheme 4). The removal of the tetrahydropyranyl ether protective group and Dess−Martin oxidation converted ether **18** to aldehyde **17** in ∼64% yield (Scheme 4), which indicates that the synthesis of aldehyde **17** via dioxolane **15a** is preferable.

The reaction of hexaconta-11Z,13Z,17Z-trien-1-al (**17**) with lithium (trimethylsilyl)acetylide,[13] presynthesized by the reaction between equimolar amounts of (trimethylsilyl)-acetylene and n-BuLi in THF, for 3 days at room temperature gave silane **20** in 91% yield. The removal of the terminal trimethylsilyl group with trimethylbutylammonium fluoride (TBAF) in THF for 4 h led to racemic lembehyne C (**21**) in a nearly quantitative yield (Scheme 5).

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For the synthesis of enantiomerically pure natural lembehyne C, racemic lembehyne C (**21**) was oxidized with the Dess–Martin periodinane (CH_2Cl_2, rt, 1 h) to give ketone **22** in 87% yield (Scheme 5). Finally, using (*R*)-2-methyl-CBS-oxazaborolidine,[14] stereoselective reduction of ketone **22** afforded natural lembehyne C (**1**) in 82% yield with high enantiomeric purity (96% ee).

Table 1. Cytotoxic Activities in Vitro of Natural Lembehyne C Measured on Tumor Cell Cultures (Jurkat, K562, HL60, U937) (nM)

compound	Jurkat	K562	U937	HL60
1	161 ± 32	373 ± 57	124 ± 19	93 ± 11

Anticancer Activity of Lembehyne C. *Cytotoxicity.* The synthesized lembehyne C (**1**) was investigated for the first time for anticancer activity in vitro using human leukemia cell lines (Jurkat, HL-60, U937 and K562). The assays included determination of IC_{50}, cell viability, effect of the compound on the cell cycle, and the mechanism of apoptosis induction. This was done using flow cytofluorimetry and Luminex xMAP multiplex assays for proteins of the key signaling pathways responsible for cell growth and proliferation.

Quantitative and qualitative analyses of cell viability, cell cycle, and apoptosis-inducing activity of lembehyne C were performed using the Guava Nexin Reagent, Guava Cell Cycle, and Guava ViaCount kits (Millipore).

The cytotoxic activity of lembehyne C in vitro against Jurkat, HL-60, U937, and K562 human leukemia cells was studied using the Guava ViaCount kit (Millipore). Similar to lembehyne B, lembehyne C showed a pronounced cytotoxicity against all types of cancer cells. However, lembehyne C had a somewhat higher cytotoxic action than lembehyne B. The highest IC_{50} value of lembehyne C was found for K562 cells (373 nM), whereas the

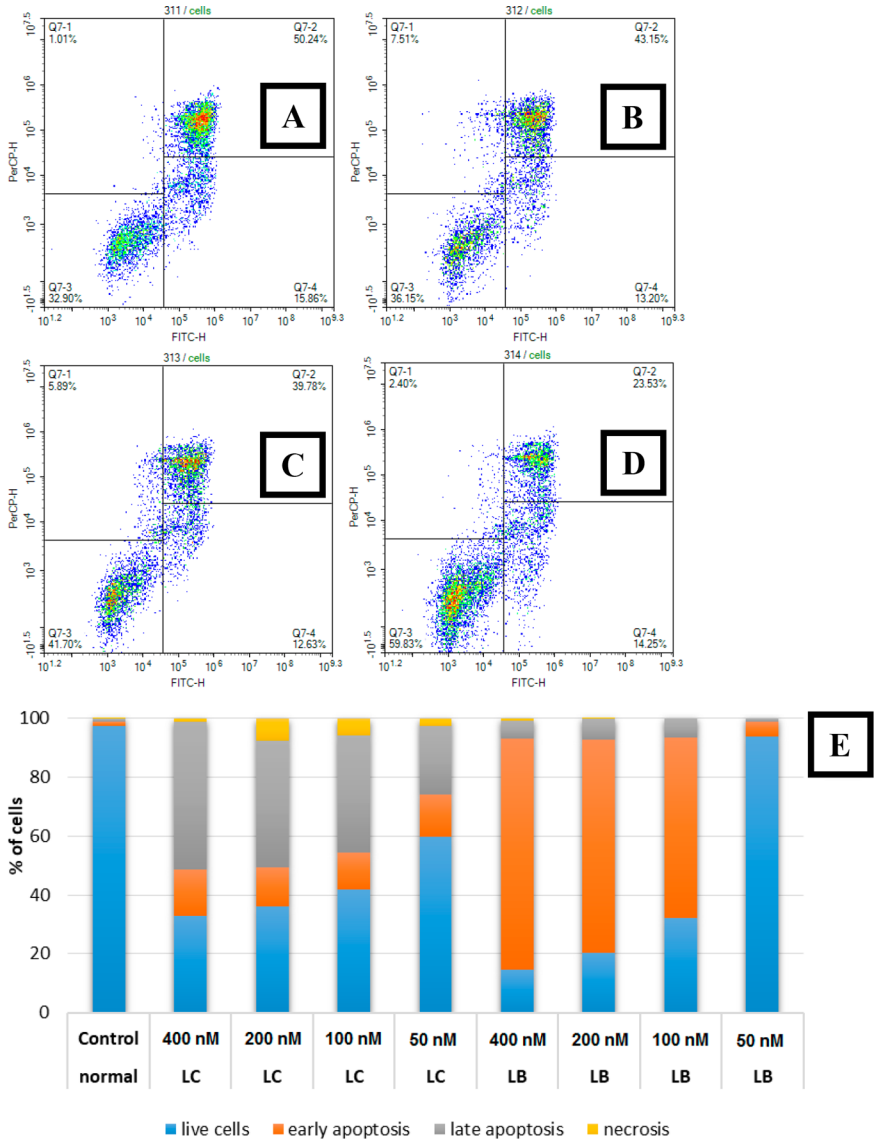


Figure 2. Cytofluorimetric analysis of the apoptosis-inducing activity of lembehyne C in Jurkat cancer cells under the action of the test compounds in various concentrations. (A) 400 nM; (B) 200 nM; (C) 100 nM; (D) 50 nM; (E) histogram of the phases of apoptosis for Jurkat cells treated with lembehyne C in various concentrations (for comparison, data for lembehyne B are given). The cells were stained with annexin V/7AAD. The incubation time was 24 h. The histogram shows the apoptosis rate (%); $^*P < 0.05$ when compared with the control group ($n = 5$).

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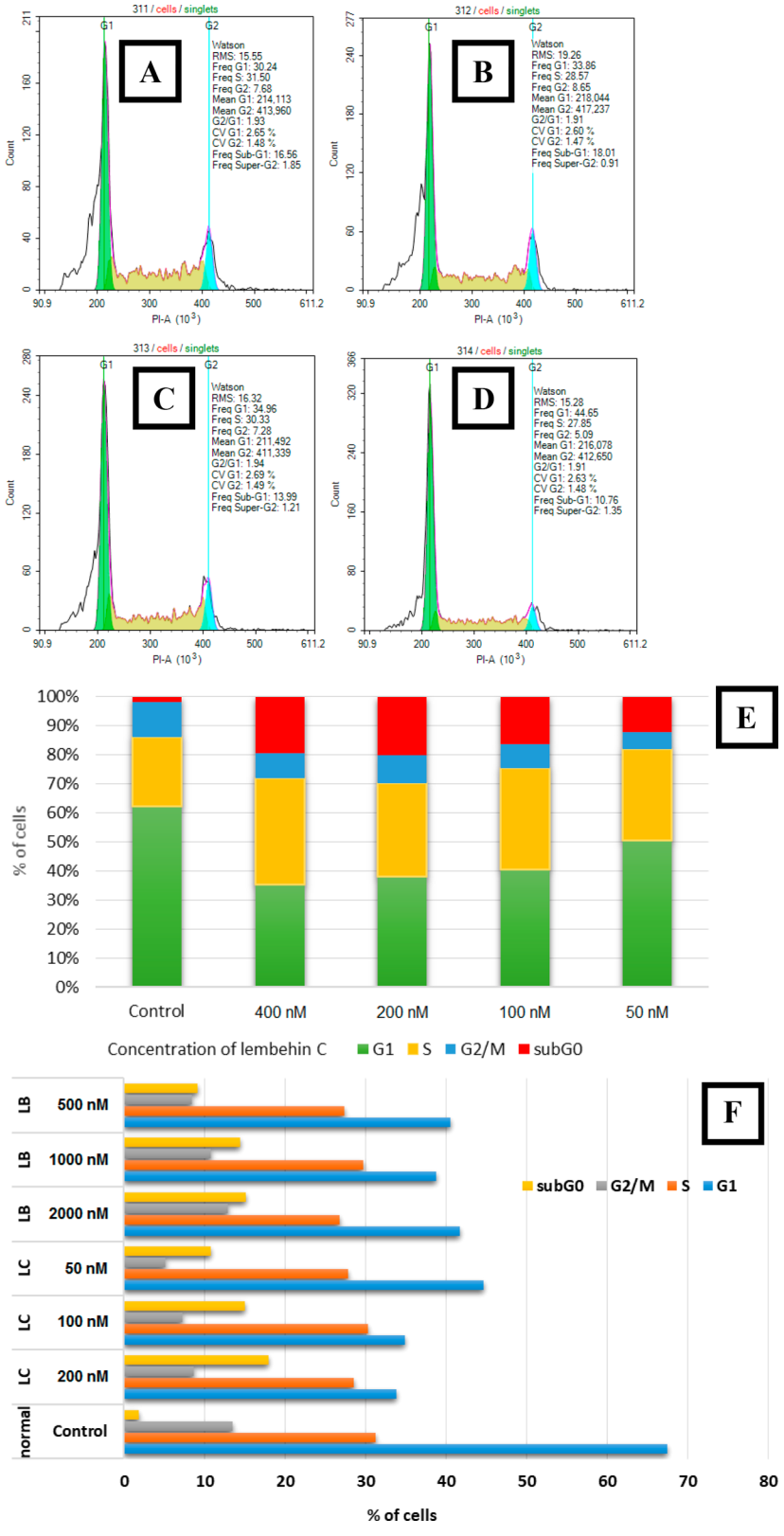


Figure 3. Cell cycle phases for Jurkat cells treated with lembehyne C **1** (LC) in various concentrations: (A) 400 nM; (B) 200 nM; (C) 100 nM; (D) 50 nM; (E) diagram of the cell cycle phases; (F) histogram of the cell cycle phases for Jurkat cells treated with lembehyne C in various concentrations [data for lembehyne B (LB) are given for comparison].[2] The incubation time of lembehynes C and B with the cells was 24 h.

values for U937, HL-60, and Jurkat IC_{50} were 124, 93, and 161 nM, respectively.

The lembehyne C-induced apoptosis in Jurkat, HL-60, U937, and K562 cells was investigated by the versatile method involving annexin V-based detection of phosphatidylserine externalization on the plasmatic membrane after treatment of cell cultures with the test compound. The effect of lembehyne C on the induction of apoptosis is more pronounced in Jurkat cells

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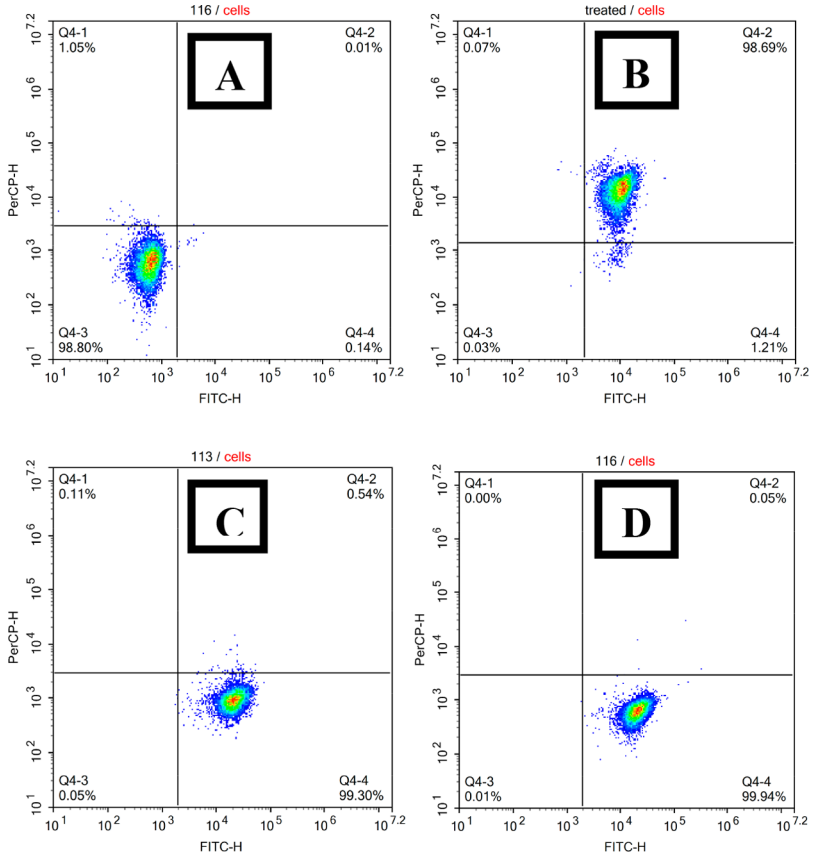


Figure 4. Detection of accumulation of phosphorylated H2A.X in Jurkat cells treated with lembehynes B and C. Control (intact living cells) (A); cells treated with etoposide in 0.2 μM concentration (B); cells treated with lembehyne C (**1**) in 0.4 μM concentration (C); cells treated with lembehyne B in 2 μM concentration (D). The incubation with test compounds lasted for 2 h.

than in other types of cancer cells, which is in line with high cytotoxicity of the compound against this cell line. As it is known from literature sources, a specific feature of Jurkat cells is the presence of a defective TP53 gene, which encodes the tumor suppressor p53 involved in maintenance of the genome stability. Thus, Jurkat cells are more stable to some apoptosis-inducing agents, which initiate apoptosis by the p53-dependent mechanism.[15] As can be seen from Figure 3, treatment of Jurkat cells with lembehyne C leads to a more pronounced dose-dependent increase in the number of late apoptotic cells than observed for lembehyne B. The highest percentage of early apoptosis (78.6%) is observed when the concentration of lembehyne B is 200 nM, whereas even at a significantly lower concentration of lembehyne C (400 nM), a several fold lower percentage of early apoptosis (15.86%) results, with late apoptosis predominating (50.24%).

According to cell cycle data determined by Guava Cell Cycle Reagent, lembehyne C is a potent inducer of hypodiploid cell population (sub-G1 phase) in all three cell lines after the appropriate treatment with the test compounds. For comparison, the hypodiploid cell population was somewhat higher for Jurkat cells at similar concentrations particularly in the case of lembehyne C: 18.1% (200 nM) (15.1% for 2 μM lembehyne B). Virtually the same trends were observed for other cancer cell suspension cultures (HL-60, K562, and U937). Thus, lembehyne C, like lembehyne B, is an apoptosis inducer in the given cell lines. Moreover, in the p53-deficient Jurkat cells, apoptosis develops in response to the treatment with lembehyne C when incubated for 24 h.

For more in-depth understanding of the mechanisms of action of lembehynes B and C, the possible DNA damage in Jurkat cells was studied. The resistance to anticancer agents that induce p53-dependent apoptosis in Jurkat cells may be caused by not only the absence of proper p53 protein or the presence of defective p53 but also by damage of other molecular targets involved in signal transduction cascade or terminal stages of apoptosis.

Simultaneous detection of phosphorylated and nonphosphorylated H2A.X protein in the same cell sample is a sensitive method for investigating the relationship between DNA damage, cell cycle checkpoints, and initiation of apoptosis. Many chemotherapeutic agents, for example, topoisomerase II inhibitors (etoposide), damage tumor cells by generating DNA double-strand breaks. It was also demonstrated that the γ-H2A.X level detected by cytofluorimetry is correlated with DNA breaks and tumor cell death.[16] With increasing DNA damage, the level of phosphorylated histone H2A.X (γH2AX) increases, particularly at DNA breaks. The H2A.X protein also plays an important role in DNA repair processes.[17]

Stimulation of Jurkat cells with the control compound, that is, etoposide, a topoisomerase II inhibitor, for 2 h results in a pronounced increase in the phosphorylation level of H2A.X (Figure 4, histogram B). Analysis of Jurkat cells treated with lembehynes B and C demonstrated that these compounds do not affect H2A.X phosphorylation (Figure 4, histograms C and D). It is quite obvious that lembehyne C, like lembehyne B, does not initiate accumulation of DNA breaks followed by activation of p53-dependent apoptosis.

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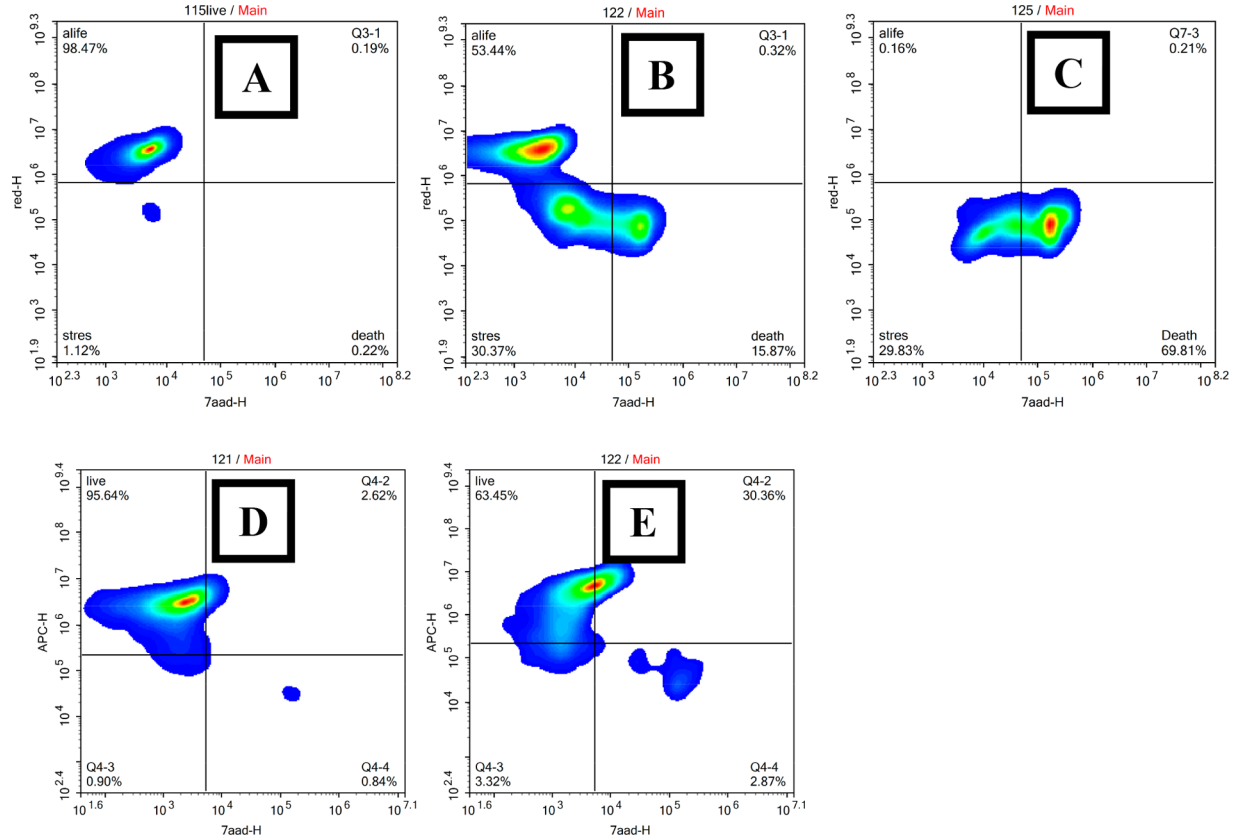


Figure 5. Detection of changes in the mitochondrial membrane potential ($\Delta\Psi$) in Jurkat cells treated with lembehyne C in different concentrations. (A) Control sample containing living cells; (B) cells treated with staurosporine (200 nM); (C) dead cells (10 μL of H_2O_2); (D) lembehyne C (**1**) (200 nM); (E) **1** (400 nM). The cells were stained with MitoSense Red and 7-AAD. The incubation time was 4 h.

Next, the possible participation of lembehyne C in activation of the mitochondrial apoptosis pathway was analyzed. For this purpose, the effect of lembehyne C on the mitochondrial membrane potential ($\Delta\Psi$) and formation of reactive oxygen species (ROS) in mitochondria were investigated. The mitochondrial regulations of apoptosis and oxidative stress are closely interrelated, which was repeatedly demonstrated in numerous studies. Detection of the mitochondrial stress and a change in the mitochondrial potential ($\Delta\Psi$) in combination with the expression of phosphatidylserine on the cell surface are considered to be reliable signs of cancer cell apoptosis.

Estimation of the mitochondrial membrane potential ($\Delta\psi$) revealed a statistically significant increase in the Jurkat cell population with a decreased membrane potential in samples treated with lembehyne C, with the change in the $\Delta\psi$ level being correlated with the test compound dose (Figure 5, histogram groups D and E). Lembehyne C decreases the mitochondrial potential of cancer cells to a minor extent; however, this effect is dose-dependent and resembles the action of staurosporine, a known inhibitor of most protein kinases (Figure 5, histogram B). The percentage of apoptotic cells in lembehyne C-treated (400 nM) samples is 6.19% (3.32% of early apoptosis and 2.87% of late apoptosis) (Figure 5E). The results attest to activation of the mitochondrial apoptosis pathway by lembehyne C via pronounced suppression of the oxidation and phosphorylation in the mitochondria of the Jurkat cells. It is clear that upon longer incubation of lembehyne C with the cancer cells, higher early and late apoptosis values can be obtained, due to a change in the potential difference of the mitochondrial membrane.

The highest oxidative stress (25.40%) induced in mitochondria by lembehyne C was detected for the cells incubated for 2 h (Figure 6B). For the 4 h incubation time, lembehyne C proved to be a much more active inducer of the oxidative stress and mitochondrial damage than classical apoptosis inducers such as etoposide (Figure 6D) or camptothecin (Figure 6E) (13.10% and 1.92%, respectively). Longer incubation (for 24 h) (Figure 6C) leads to a somewhat different situation: most of the cells successively switch to apoptosis (45.78%) via the stage of mitochondrial damage (20.58%).

To continue the study of the anticancer activity of lembehyne C, the protein profiles of the most universal signaling cascades such as CREB, JNK, NFkB, p38, ERK1/2, Akt, p70S6K, STAT3, and STAT5, responsible for coordinated regulation of the transcription and translation apparatus of the cancer cell and, hence, responsible for the growth and protein synthesis function, which appears especially important for evaluation of tumor progression, particularly for the p53-deficient cancer cells, were investigated. The key goal of the investigation of the above-mentioned signaling cascades was to elucidate the effect of lembehyne C on the cell cycle and the mechanism of the lembehyne C anticancer action.

Generally, considering the pairwise comparison of phosphorylated and nonphosphorylated fractions of proteins, the major changes for the ratio between these two fractions occur for the Akt, p38, CREB, and p70S6K signaling pathways (Figure 7A). Signaling via RTKs (tyrosine kinase receptor) often enhances metabolism and cell growth. ERK/MAP and Akt are two key Ser/Thr family kinases activated by the RTK signaling, which increase the activity of the p70 S6, Msk1, STAT3

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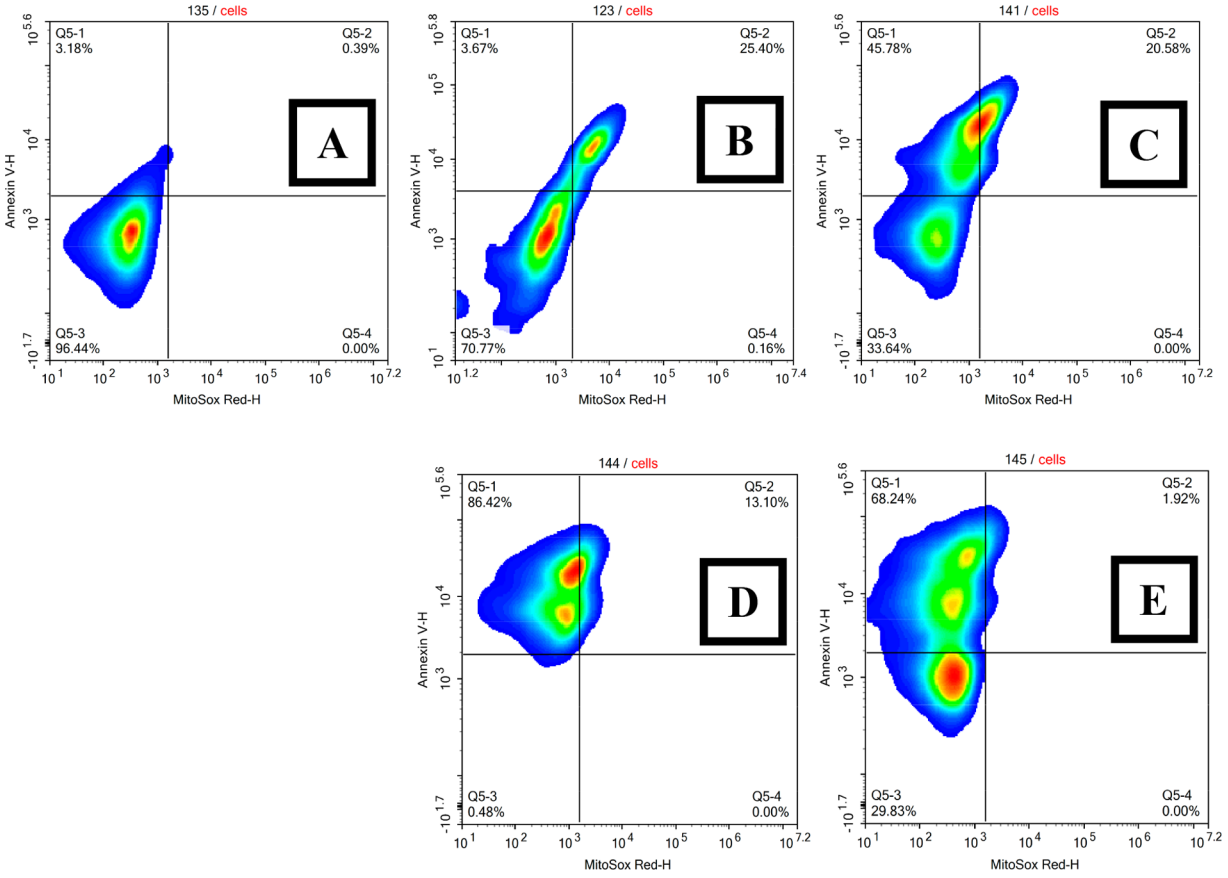


Figure 6. Generation of reactive oxygen species and apoptotic cell death in Jurkat cells treated with lembehyne C in different concentrations and with different incubation times: (A) control sample; (B) lembehyne C (**1**) (200 nM), 4 h incubation; (C) (**1**) (400 nM) 24 h incubation; (D) etoposide (200 nM), 4 h incubation; (E) camptothecin (200 nM), 4 h incubation. The cells were stained with MitoSox Red and annexin V-CF488A. Q5-1, apoptotic cells; Q5-2, both apoptotic cells and cells with clear mitochondrial damages; Q5-3, living undamaged cells; Q5-4, cells with damaged mitochondria.

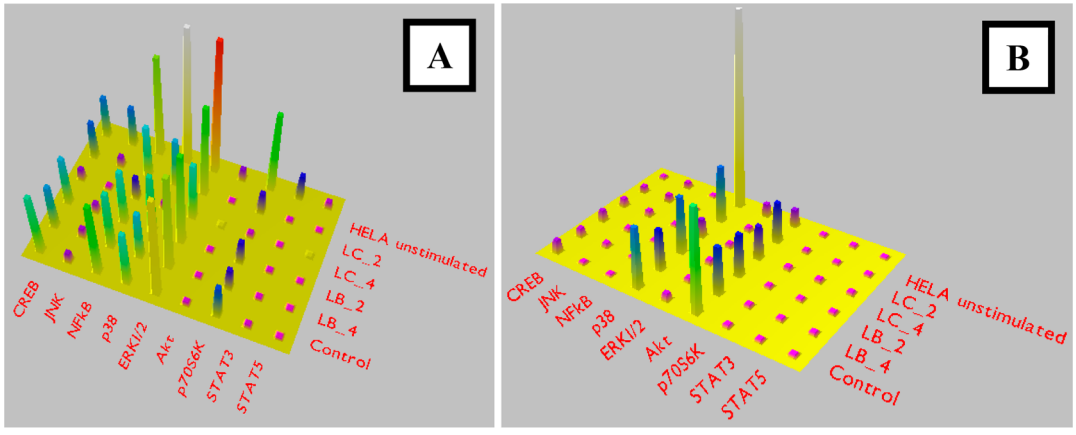


Figure 7. Histograms of the major signaling pathways in Jurkat tumor cells treated with lembehynes C (LC, **1**) and B (LB). Total (A) and phosphorylated (B) CREB, JNK, NFkB, p38, ERK1/2, Akt, p70S6K, STAT3, and STAT5 proteins were determined. Lembehyne C (**1**) was used in concentrations of 200 nM (LC_2) and 400 nM (LC_4); lembehyne B was taken in concentrations of 1000 nM (LB_2) and 2000 nM (LB_4). The images were drawn using xPONENT 4.2 for MAGPIX software.

(Ser727), and CREB kinases and activate other intermediates. Other signaling pathways induced by stress or death receptor cause activation of p38, JNK, and NF-κB. As can be seen from Figure 7(A), lembehyne C in 400 nM concentration causes the most pronounced decrease for all types of protein kinases in a cancer cell. Lembehyne B (2 μM) is somewhat inferior to lembehyne C in activity; however, it also efficiently influences the contents of phosphorylated forms of various kinases in comparison with the control.

In conclusion, we developed the first original *Z*-stereoselective total synthesis of a natural alkynol, lembehyne C (**1**), with the new Ti-catalyzed cross-cyclomagnesiation of oxygenated and aliphatic 1,2-dienes with available Grignard reagents as the key step. Also, the anticancer properties of

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lembehyne C were studied for the first time. The inhibition of the main kinases of four signaling pathways by lembehyne C has obviously a suppressing effect on the growth and proliferation of cancer cells, even in the case of a p-53-deficient cancer cell line. Lembehyne C, being an efficient inducer of apoptosis activated via the mitochondrial pathway and an inhibitor of a number of important signaling pathways of cell growth and proliferation, may become a drug candidate with a high anticancer potential.

EXPERIMENTAL SECTION

General Experimental Procedures. Melting points were recorded on a Stuart SMP3 apparatus. IR spectra were recorded on a Bruker Vertex 70 V spectrometer using KBr discs over the range of 400−4000 cm^{-1}. ^1H and ^{13}C NMR spectra were obtained using a Bruker Ascend 500 spectrometer in $CDCl_3$ operating at 500 MHz for ^1H and 125 MHz for ^{13}C and a Bruker AVANCE 400 spectrometer in $CDCl_3$ operating at 400 MHz for ^1H and 100 MHz for ^{13}C. Mass spectra were obtained with a MALDI TOF/TOF spectrometer in an α-cyano-4-hydroxycinnamic acid matrix. HR-ESIMS data were measured on a Bruker maXis instrument using electrospray ionization. In experiments on selective collisional activation (CAD) the activation energy was set at maximum abundance of fragment peaks (see figure legends). A syringe injection was used for solutions in MeCN/H_2O, 50/50 vol % (flow rate 3 mL/min). Nitrogen was applied as a dry gas; the interface temperature was set at 180 °C. Elemental analyses were measured on a 1106 Carlo Erba apparatus. The purity of the synthesized compounds was controlled using TLC on Sorbfil plates; anisic aldehyde in HOAc was used for color development. Column chromatography was carried out on Acrus silica gel (0.060−0.200 mm). All solvents were dried (1,4-dioxane, THF, Et_2O over Na) and freshly distilled before use.

*Synthesis of Icos-3-yn-1-ol (**5**).* To a solution of octadec-1-yne (20 g, 80 mmol) in THF (50 mL) was slowly added EtMgBr (3.0 M in THF, 32.0 mL, 96.0 mmol). The mixture was heated under reflux for 4 h and cooled over ice. Ethylene oxide (5.85 g, 130 mmol) was decanted into ice cold dry THF (30 mL) and slowly added to the cooled Grignard reagent. A slight exothermic reaction occurred. The mixture was allowed to warm to room temperature, stirred overnight prior to cooling over ice, and hydrolyzed by careful addition of HCl (50 mL 5%). The mixture was extracted with EtOAc (2 × 100 mL), dried over $MgSO_4$, and evaporated to give **5** (16.0 g, 68%) as a colorless oil: ^1H NMR (400 MHz, $CDCl_3$) δ 3.63 (t, J = 6.4 Hz, 2H), 2.64 (s, 1H), 2.40−2.36 (m, 2H), 2.19−2.07 (m, 2H), 1.47−1.42 (m, 2H), 1.39−1.23 (m, 26H), 0.85 (t, J = 6.7 Hz, 3H); ^{13}C NMR (100.62 MHz, $CDCl_3$) δ 82.32, 76.31, 61.27, 31.92, 29.70, 29.55, 29.37, 29.17, 29.01, 28.91, 23.04, 22.67, 18.70, 14.04; HRMS (ESI-TOF) calcd for $C_{20}H_{38}O$ $[M + H]^+$ 295.3000, found 295.3006. Anal. Calcd for $C_{22}H_{38}O$: C, 81.55; H, 13.01. Found: C, 81.65; H, 13.12.

*Synthesis of (Z)-Icos-3-en-1-ol (**6**).* To a stirred mixture of $Ni(OAc)_2{\cdot}4H_2O$ (10.7 g, 42.7 mmol) in EtOH (86 mL) was added $NaBH_4$ (1.2 g, 31 mmol) in EtOH (43 mL) at room temperature under hydrogen. The mixture was stirred for 60 min, and ethylenediamine (10.32g, 171.5 mmol) and alkynol **5** (12.6 g, 43 mmol) were added sequentially. After stirring for 12 h at the same temperature under hydrogen, the reaction mixture was concentrated and diluted with EtOAc. The resulting mixture was filtered through a Celite pad, and the filtrate was concentrated under reduced pressure. The residue was purified by silica gel chromatography (*n*-hexane/EtOAc, 5:1) to furnish **6** (12.24 g, 96% yield) as a colorless oil: ^1H NMR (400 MHz, $CDCl_3$) δ 5.59−5.53 (m, 1H), 5.40−5.33 (m, 1H), 3.64 (t, J = 6.6 Hz, 2H), 2.33 (q, J = 6.6 Hz, 2H), 2.09−1.96 (m, 2H), 1.27 (m, 28H), 0.89 (t, J = 6.7 Hz, 3H); ^{13}C NMR (100.62 MHz, $CDCl_3$) δ 133.40, 124.94, 62.26, 31.94, 30.77, 29.72, 29.57, 29.39, 29.34, 27.38, 22.70, 14.11; HRMS (ESI-TOF) calcd for $C_{20}H_{40}O$ $[M + H]^+$ 297.3157, found 297.3152. Anal. Calcd for $C_{20}H_{40}O$: C, 80.99; H, 13.60. Found: C, 80.94; H, 13.65.

*Synthesis of (Z)-1-Bromoicos-3-ene (**8**).* A mixture of CH_3SO_2Cl (4.93 g, 43.0 mmol), triethylamine (8.22 mL), (*Z*)-icos-3-en-1-ol (**6**) (10.0 g, 34.0 mmol), and CH_2Cl_2 (52.2 mL) was stirred at 0 °C for 1 h. The reaction mixture was filtered and the residue washed with CH_2Cl_2 (40 mL). The combined CH_2Cl_2 layer was washed with diluted HCl (30 mL) and water (2 × 25 mL) and dried (Na_2SO_4), and the solvent was removed under reduced pressure. To a solution of **7** (11.88 g, 94%) in dry acetone (104 mL) was added LiBr (6.0 g, 69.2 mmol), and the mixture refluxed for 3 h. Ice cold water (60 mL) was added, and the oil that separated was removed. The aqueous layer was extracted with *n*-hexane (2 × 50 mL). The *n*-hexane extract was combined with the oily liquid, washed with water (20 mL), and dried over anhydrous Na_2SO_4. Evaporation of the solvent, followed by column chromatography of the residue on silica gel, using *n*-hexane/Et_2O (95:5) as eluent, gave **8** (9.8 g, 85%) as a yellow liquid oil: ^1H NMR (400 MHz, $CDCl_3$) δ 5.59−5.52 (m, 1H), 5.41−5.35 (m, 1H), 3.38 (t, J = 7.2 Hz, 2H), 2.64 (q, J = 7.1 Hz, 2H), 2.06 (q, J = 7.2 Hz, 2H), 1.38−1.28 (m, 28H), 0.90 (t, J = 6.7 Hz, 3H); ^{13}C NMR (100.62 MHz, $CDCl_3$) δ 133.21, 125.72, 32.52, 31.95, 30.87, 29.72, 29.64, 29.54, 29.39, 29.30, 27.44, 22.71, 14.12; HRMS (ESI-TOF) calcd for $C_{20}H_{39}{}^{79}Br$ $[M + H]^+$ 359.2313, found 359.2315. Anal. Calcd for $C_{20}H_{39}Br$: C, 66.99; H, 10.97. Found: C, 66.93; H, 10.98.

*Synthesis of Docos-5-en-1-yne (**9**).* To a suspension of a lithium acetylide/ethylenediamine complex (3.1 g, 36.9 mmol) in DMSO (13.6 mL) was added **8** (8.0 g, 22.2 mmol) in DMSO (11.2 mL) at 0 °C. The reaction mixture was stirred for 12 h at 23 °C, saturated NH_4Cl (10 mL) was added, and the mixture was extracted with Et_2O, washed with brine, dried over $MgSO_4$, and concentrated. The residue was chromatographed over silica gel (*n*-hexane) to afford **9** (4.5 g, 67%) as a colorless oil: ^1H NMR (400 MHz; $CDCl_3$) δ 5.49−5.37 (m, 2H), 2.28 (m, 2H), 2.22 (m, 2H), 2.04 (q, J = 6.8 Hz, 2H), 1.94 (t, J = 2.4 Hz, 1H), 1.39−1.26 (m, 28H), 0.88 (t, J = 6.7 Hz, 3H); ^{13}C NMR (100.62 MHz; $CDCl_3$) δ 131.7, 127.4, 84.2, 68.2, 31.9, 29.7−29.3, 27.3, 26.4, 22.7, 18.9, 14.1; HRMS (ESI-TOF) calcd for $C_{22}H_{40}$ $[M + H]^+$ 305.3208, found 305.3202. Anal. Calcd for $C_{22}H_{40}$: C, 86.75; H, 13.24. Found: C, 86.77; H, 13.29.

*Synthesis of (Z)-Tricosa-1,2,6-triene (**10**).* Paraformaldehyde (1.0 g), CuI (1.26 g, 6.8 mmol), and dicyclohexylamine (75.7 mL) were sequentially added to a solution of **9** (5.0 g, 16.4 mmol) in anhydrous dioxane (75.7 mL). The resulting mixture was heated under reflux for 8 h, the solvent was evaporated in vacuo, 2 M HCl (50 mL) was added, and the mixture was extracted with Et_2O. The ethereal solution was washed with $NaHCO_3$, water, and brine and dried with anhydrous $MgSO_4$, and the residue was purified by silica gel column chromatography (*n*-hexane/EtOAc, 30:1) to afford **10** (3.96 g, 76%) as a colorless liquid: ^1H NMR (400 MHz, $CDCl_3$) δ 5.50−5.40 (m, 2H), 5.22−5.09 (m, 1H), 4.69 (m, 2H), 2.23−2.16 (m, 2H), 2.10−2.03 (m, 4H), 1.46−1.20 (m, 28H), 0.91 (t, J = 6.7 Hz, 3H); ^{13}C NMR (100.62 MHz, $CDCl_3$) δ 208.51, 130.77, 128.63, 89.64, 74.80, 31.95, 29.72, 29.63, 29.58, 29.52, 29.38, 29.33, 29.26, 27.29, 26.85, 22.71, 14.12; HRMS (ESI-TOF) calcd for$C_{20}H_{38}O$ $[M + H]^+$ 319.3364, found 319.3361. Anal. Calcd for $C_{22}H_{38}O$: C, 86.70; H, 13.29. Found: C, 86.66; H, 13.28.

*Cross-Cyclomagnesiation of (Z)-Tricosa-1,2,6-triene (**10**) and 2-(Dodeca-10,11-dien-1-yl)-1,3-dioxolane (**15a**) or 2-(Dodeca-10,11-dien-1-yloxy)tetrahydro-2H-pyran (**15b**) with EtMgBr in the Presence of Mg Metal and Cp_2TiCl_2 Catalyst (General Procedure).* Diethyl ether (30 mL), (*Z*)-tricosa-1,2,6-triene (**10**) (1.0 g, 4.8 mmol), 2-(dodeca-10,11-dien-1-yl)-1,3-dioxolane (**15a**) or 2-(dodeca-10,11-dien-1-yloxy)tetrahydro-2*H*-pyran (**15b**) (4.0 mmol), EtMgBr (40.0 mmol) (as a 1.5 M solution in Et_2O), Mg powder (0.7 g, 30.0 mmol), and Cp_2TiCl_2 (0.1 g, 0.4 mmol) were placed in a glass reactor with stirring under argon (~0 °C). The mixture was warmed to room temperature (20−22 °C) and stirred for 6 h. The reaction mixture was treated with a 5% solution of NH_4Cl in H_2O (20 mL) and extracted with Et_2O (2 × 100 mL). The combined organic phases were dried over $MgSO_4$ and filtered, and the solvents were removed under reduced pressure. Silica gel column chromatography (*n*-hexane/EtOAc, 35:1) of the residue gave compound **17** (1.8 g, 81%) or **18** (2.14 g, 89%) as a pale yellow oily liquid.

*(11Z,15Z,19Z)-Hexatriaconta-11,15,19-trienal (**17**):* IR ν_{max} 3007, 2922, 2852, 1729, 1466, 1091, 910, 722 cm^{-1}; ^1H NMR (400 MHz,

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$CDCl_3$) δ 9.78 (s, 1H), 5.40−5.38 (m, 6H), 2.43 (td, J = 7.4, 1.5 Hz, 2H), 2.10−2.02 (m, 12H), 1.66−1.61 (m, 2H), 1.32−1.28 (m, 40H), 0.90 (t, J = 6.7 Hz, 3H); ^{13}C NMR (100.62 MHz, $CDCl_3$) δ 202.82, 130.41, 130.32, 129.60, 129.13, 129.07, 43.91, 31.93, 29.71, 29.67, 29.58, 29.48, 29.40, 29.37, 29.28, 29.17, 27.45, 27.38, 27.27, 22.69, 22.09, 14.11; HRMS (ESI-TOF) calcd for $C_{36}H_{66}O$ $[M + H]^+$ 515.5191, found 515.5194. Anal. Calcd for $C_{36}H_{66}O$: C, 83.96; H, 12.92. Found: C, 83.93; H, 12.89.

*2-[(11Z,15Z,19Z)-Hexatriaconta-11,15,19-trien-1-yloxy]-tetrahydro-2H-pyran (**18**):* ^{1}H NMR (400 MHz, $CDCl_3$) δ 5.40−5.38 (m, 6H), 4.59 (t, J = 3.2 Hz, 1H), 3.91−3.36 (m, 4H), 2.10−2.03 (m, 12H), 1.86−1.28 (m, 52H), 0.90 (t, J = 7.2 Hz, 3H); ^{13}C NMR (100.62 MHz, $CDCl_3$) δ 130.35, 129.58, 129.05, 98.76, 67.64, 62.20, 31.96, 30.78, 29.74, 29.70, 29.54, 29.41, 29.36, 27.46, 27.39, 27.28, 26.28, 25.54, 22.72, 19.67, 14.12; HRMS (ESI-TOF) calcd for $C_{41}H_{76}O_2$ $[M + H]^+$ 601.5923, found 601.5925. Anal. Calcd for $C_{41}H_{76}O_2$: C, 81.92; H, 12.75. Found: C, 81.89; H, 12.83.

THP deprotection of ether (**18**) was carried out with p-TsOH in $CHCl_3$ /MeOH using a known method[13] to afford (11*Z*,15*Z*,19*Z*)-hexatriaconta-11,15,19-trien-1-ol (**19**): IR ν_{max} 2925, 2854, 2251, 1718, 1465, 1055, 909, 735 cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$) δ 5.40−5.38 (m, 6H), 3.65 (t, J = 6.8 Hz, 2H), 2.75−2.71 (m, 1H), 2.10−2.03 (m, 12H), 1.60−1.28 (m, 44H), 0.90 (t, J = 6.8 Hz, 3H); ^{13}C NMR (100.62 MHz, $CDCl_3$) δ 130.41, 130.37, 129.61, 129.08, 63.05, 32.82, 31.94, 29.72, 29.59, 29.45, 29.34, 27.46, 27.38, 27.27, 25.76, 22.70, 14.12; HRMS (ESI-TOF) calcd for $C_{36}H_{68}O$ $[M + H]^+$ 517.5348, found 517.5345. Anal. Calcd for $C_{36}H_{68}O$: C, 83.64; H, 13.26. Found: C, 83.67; H, 13.33.

The oxidation of alcohol (**19**) with Dess−Martin periodinane was carried out according to a known procedure.[13]

*Procedure for Preparation of (15Z,19Z,23Z)-1-(Trimethylsilyl)-tetraconta-15,19,23-trien-1-yn-3-ol (**20**).* To a solution of trimethylsilyl acetylene (0.183 g, 1.87 mmol) in THF (2.13 mL) was added dropwise a solution of 0.67 mL of *n*-BuLi (2.5 M in *n*-hexane) at −78 °C. The temperature was adjusted to −20 °C, and a THF (2.13 mL) solution of 0.529 g (1.00 mmol) of dienal (**17**) was added at the same temperature. The reaction mixture was warmed to room temperature (20−22 °C) and stirred for 3 days. The mixture was treated with a 5% solution of NH_4Cl in H_2O (20 mL) and extracted with Et_2O (30 mL). The combined organic phases were dried over $MgSO_4$ and filtered, and the solvents were removed under reduced pressure. Silica gel column chromatography of the residue gave compound **20** (0.55 g, 91%) as a pale yellow solid.

*(15Z,19Z,23Z)-1-(Trimethylsilyl)tetraconta-15,19,23-trien-1-yn-3-ol (**20**).* IR ν_{max} 3307, 2927, 2855, 2253, 1681, 1466, 1383, 1096, 908, 733 cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$) δ 5.40−5.39 (m, 6H), 4.37 (dd, J = 12.1, 6.2 Hz, 1H), 2.11−2.03 (m, 12H), 1.84−1.28 (m, 44H), 0.90 (t, J = 6.7 Hz, 3H), 0.19 (s, 9H); ^{13}C NMR (100.62 MHz, $CDCl_3$) δ 130.42, 130.38, 129.62, 129.11, 106.97, 89.25, 62.91, 37.72, 31.94, 29.72, 29.68, 29.59, 29.53, 29.38, 29.34, 29.23, 27.46, 27.38, 27.28, 25.12, 22.70, 14.13, −0.11; HRMS (ESI-TOF) calcd for $C_{41}H_{76}OSi$ $[M + H]^+$ 613.5743, found 613.5745. Anal. Calcd for $C_{41}H_{76}OSi$: C, 80.31; H, 12.50. Found: C, 80.34; H, 12.53.

*Procedure for the Preparation of rac-Lembehyne C (**21**).* To a solution of alkyne (**20**) (0.55 g, 0.9 mmol) in THF (3.74 mL) was added TBAF (1 M in THF, 1.2 equv), and the solution stirred for 2 h at room temperature. The reaction mixture was treated with saturated aqueous NH_4Cl and extracted with Et_2O (2 × 10 mL). The combined organic phases were dried over $MgSO_4$ and filtered, and the solvents were removed under reduced pressure. Silica gel column chromatography of the residue gave compound **21** (0.48 g, 99%) as a colorless waxy solid.

The oxidation of the alcohol (**21**) with Dess−Martin periodinane was carried out according to a published procedure.[18]

*(13Z,17Z,21Z)-Octatriaconta-13,17,21-trien-1-yn-3-one (**22**):* IR ν_{max} 3301, 2925, 2854, 2253, 2095, 1683, 1465, 1370, 1088, 909, 846, 735, 649 cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$) δ 5.38 (m, 6H), 3.21 (d, J = 1.3 Hz, 1H), 2.60 (dd, J = 10.0, 4.7 Hz, 2H), 2.10−2.04 (m, 12H), 1.69 (m, 2H), 1.28 (m, 44H), 0.89 (t, J = 6.9, 3H); ^{13}C NMR (100.62 MHz, $CDCl_3$) δ 187.36, 130.38, 130.29, 129.60, 129.12, 129.06, 81.49, 78.13, 77.35, 77.03, 76.71, 45.44, 31.95, 29.72, 29.60, 29.47, 29.39, 29.34, 29.29, 28.90, 27.45, 27.38, 27.27, 23.75, 22.70, 14.11; HRMS (ESI-TOF) calcd for $C_{38}H_{66}O$ $[M + H]^+$ 539.5191, found 539.5193. Anal. Calcd for $C_{38}H_{66}O$: C, 84.67; H, 12.35. Found: C, 84.65; H, 12.31.

The stereoselective reduction of ketone **22** with (*R*)-2-methyl-CBS-oxazaborolidine was carried out according to a published procedure.[14]

*Lembehyne C (**1**):* colorless powder; $[\alpha]^{22}_D$ +0.5 (c 0.2, $CHCl_3$); IR ν_{max} 3462, 3301, 2925, 2853, 2252, 2151, 1683, 1464, 1253, 1087, 909, 846, 761, 649 cm^{-1}; yield 82% (96% ee); ^{1}H NMR (400 MHz, $CDCl_3$) δ 5.41−5.39 (m, 6H), 4.39 (t, J = 5.9 Hz, 1H), 2.47 (d, J = 1.8 Hz, 1H), 2.11−1.94 (m, 12H), 1.83−1.17 (m, 44H), 0.90 (t, J = 6.6 Hz, 3H); ^{13}C NMR (100.62 MHz, $CDCl_3$) δ 130.42, 130.37, 129.62, 129.12, 129.09, 85.04, 72.80, 62.33, 37.66, 31.94, 29.72, 29.68, 29.59, 29.53, 29.38, 29.35, 29.32, 29.25, 27.46, 27.39, 27.28, 25.02, 22.70, 14.12; HRMS (ESI-TOF) calcd for $C_{38}H_{64}O$ $[M + H]^+$ 541.5348, found 541.5349. Anal. Calcd for $C_{38}H_{64}O$: C, 84.36; H, 12.67. Found: C, 84.33; H, 12.01.

Biology. *Cell Culturing.* Cells (Jurkat, K562, U937, HL60) were purchased from HPA Culture Collections (Salisbury, UK) and cultured according to standard mammalian tissue culture protocols and sterile technique. All cell lines used in the study were tested and shown to be free of mycoplasma and other viral contamination. Cells were maintained in RPMI 1640 (Jurkat, K562, U937, HL60) (Gibco) supplemented with 4 mM glutamine, 10% FBS (Sigma), and 100 units/mL penicillin−streptomycin (Sigma). All types of cells were grown in an atmosphere of 5% CO_2 at 37 °C. The cells were subcultured at 2−3-day intervals. Jurkat, K562, U937, and HL60 cells were subcultured at 2-day intervals with a seeding density of 1 × 10^5 cells per 24-well plate in RPMI with 10% FBS.

Cytotoxicity Assay. Viability (live/dead) assessment was performed by staining cells with 7-AAD (7-aminoactinomycin D) (Biolegend). After treatment cells were harvested, washed one or two times with phosphate-buffered saline (PBS), and centrifuged at 400*g* for 5 min. Cell pellets were resuspended in 200 mL of flow cytometry staining buffer (PBS without Ca^{2+} and Mg^{2+}, 2.5% FBS) and stained with 5 μL of 7-AAD staining solution for 15 min at room temperature in the dark. Samples were acquired on NovoCyte 2000 flow cytometry system (ACEA) equipped with a 488 nm argon laser. Detection of 7-AAD emission was collected through a 675/30 nm filter in FL4 channel.

Viability and Apoptosis. Apoptosis was determined by flow cytometric analysis of annexin V and 7-AAD staining. After treatment cells during 24 h were harvested, washed one or two times with PBS, and centrifuged at 400*g* for 5 min. Cell pellets were resuspended in 200 μL of flow cytometry staining buffer (PBS without Ca^{2+} and Mg^{2+}, 2.5% FBS). Then, 200 μL of Guava Nexin reagent (Millipore, Bedford, MA, USA) was added to 5 × 10^5 cells in 200 μL, and the cells were incubated with the reagent for 20 min at room temperature in the dark. At the end of incubation, the cells were analyzed on a NovoCyte 2000 flow cytometry system (ACEA).

Cell Cycle Analysis. The cell cycle was analyzed using the method of propidium iodide staining. After treatment, cells during 24 h were harvested, washed one or two times with PBS, and centrifuged at 400*g* for 5 min. Cell pellets were resuspended in 200 μL of flow cytometry staining buffer (PBS without Ca^{2+} and Mg^{2+}, 2.5% FBS). Cells were plated in 24-well round-bottom plates at a density 10 × 10^5 cells per well, centrifuged at 450*g* for 5 min, and fixed with ice-cold 70% EtOH for 24 h at 0 °C. Cells were washed with PBS and incubated with 250 μL of Guava Cell Cycle Reagent (Millipore) for 30 min at room temperature in the dark. Samples were analyzed on NovoCyte 2000 flow cytometry system (ACEA).

Histone H2A.X Analysis. Cells for phosphorylation of histone H2A.X in Jurkat cell analysis were grown in a culture flask for 48 h. The medium was changed 24 h before drug treatment. Phosphorylation of histone H2A.X was measured with the FlowCellect DNA damage histone H2A.X dual detection kit (FCCS025153, Millipore). Briefly, isolated and treated cells were counted and diluted with assay buffer (5 × 10^6 cells/2.5 mL). Paraformaldehyde (PFA, Sigma) was added at a final concentration of 2%, and cells were fixed at 37 °C for 10 min. Cells were immediately chilled on ice, centrifuged, washed three times with 0.5%

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bovine serum albumin (BSA; EMD Bioscience, La Jolla, CA, USA)/PBS, permeabilized with 70% EtOH in distilled water or 90% MeOH in 0.5% BSA/PBS, and kept at 4 °C for EtOH samples and at −20 °C for MeOH samples until further staining. After that, the cells were fixed, permeabilized, incubated with anti-phosphohistone H2A.X (Ser139, Alexa Fluor555, part number CS208203, Millipore) and anti-H2A.X (PECy5, part number CS209202, Millipore), and then analyzed by flow cytometry.

Mitochondrial Damage. A cytometric assay, which allowed multiparametric evaluation of three cell health markers—change in mitochondrial potential (early apoptosis and cellular stress), phosphatidylserine expression on the cell surface (late apoptosis), and membrane permeabilization (cell death)—was performed. Millipore's FlowCellectMitoDamage kit use of the reagents allows researchers to obtain information on early and late apoptosis in one simple assay. Cells were treated with drugs at a concentration of 0.1−0.4 μM and incubated (37 °C, 5% CO_2) for 4 h. After this time, the cells were dissociated with Accutase solution, stained, and analyzed by flow cytometry (NovoCyte flow cytometry) according to the manufacturer's protocols (FlowCellect MitoDamage kit, Merck, and FlowCellect oxidative stress characterization kit, Merck).

Multiplex Analysis of Early Apoptosis Markers. The Milliplex MAP 9-Plex multi-pathway kits, total, are used to detect total protein levels or phosphorylated protein levels for ERK/MAP kinase 1/2, Akt, STAT3, JNK, p70 S6 kinase, NFκB, STAT5A/B, CREB, and p38 in cell lysates using the Luminex system. The detection assay is a rapid, convenient alternative to Western blotting and immunoprecipitation procedures. Each kit has sufficient reagents for one 96-well plate assay. Multiplexed immunoassays 9-Plex multi-pathway total magnetic bead kit 96-well plates from Millipore were used to detect changes in phosphorylated cAMP response element-binding protein (CREB; pS133), ERK (pT185/pY187), NF-kB (pS536), JNK (pT183/pY185), p38 (pT180/pY182), p70 S6K (pT412), STAT3 (pS727), STAT5A/B (pY694/699), and Akt (pS473) (Milliplex 48-680MAG; Merck Millipore, Germany) in Jurkat cell lysates lysed with RIPA buffer and protease and phosphatase inhibitors according to the array protocol. Before lysis in Milliplex MAP lysis buffer, the cells were challenged for 3 min with 100 ng/mL EGF and 10 nM HRG-b1 and prepared for subsequent analysis.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jnatprod.0c00261.

NMR data, NMR spectra, and general procedure (PDF)

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Notes

The authors declare no competing financial interest.

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